

Global Vaccine-Preventable Disease Surveillance Dashboard

Interactive Excel dashboard of WHO-reported cases of vaccine-preventable diseases, 1980-2025.

Question

How have reported cases of each disease changed over time, which countries and regions report the most, and how much of a trend could be a reporting artifact?

Data

- Source: WHO Immunization Data Portal (reported cases, WHO/UNICEF Joint Reporting Form)
- Downloaded: 28 Sep 2026
- 13 diseases, 217 countries, 6 WHO regions, 1980-2025
- About 23% of case cells are blank (not reported). Blanks are never treated as zero.

What I did

1. Audited my first version and found a double-counting error: the trend chart summed COUNTRIES + WHO_REGIONS + GLOBAL rows, inflating totals about 3x (1980 total tetanus showed 341,833; the correct country-level figure is 113,961).
2. Removed a junk footer row, corrected the title year (data ends 2025), and made the disease an explicit dropdown (the old version was silently filtered to tetanus).
3. Rebuilt the workbook with formula-driven views: disease, from-year and to-year dropdowns drive the charts and the written findings.
4. Added a "countries reporting" chart, because a change in how many countries report can look like a change in disease.

Findings (Total tetanus view)

- Reported cases fell from 113,961 (1980) to 27,200 (2025), about -76%, with a low near 10,000 around 2015 followed by a rise.
- The recent rise coincides with more countries reporting (148 in 2015, 185 in 2025), so it should not be read as a pure increase in disease.
- India has the highest cumulative reported cases, but raw counts are not adjusted for population.

Limitations

- No population adjustment (per-100,000 rates would need UN population data).
- Reporting completeness varies by country and era.
- Descriptive only; no causal claim about vaccination.

Files

- `Vaccine-Preventable-Disease-Dashboard.xlsx` : Dashboard, Notes (method and fixes), Calc, Data

Tools

Excel (formulas, charts, data validation); Python (pandas) for the audit.